MIP_LINEAGE_MODEL.md

Mainframe Intelligence Platform

Enterprise Data & Execution Lineage Architecture

Version: 1.0

Status: Core Intelligence Layer

Classification: Strategic Platform Component

Purpose

Lineage explains how information and execution flow through a legacy system.

Lineage enables MIP to answer:

- Where did this data originate?
- How did it get here?
- What programs transformed it?
- What jobs consumed it?
- What systems depend on it?
- What breaks if it changes?

Without lineage:

```
Metadata → Static Understanding
```

With lineage:

```
Metadata
    ↓
Relationships
    ↓
Lineage
    ↓
Impact Analysis
    ↓
Modernization Intelligence
```

Executive Vision

A modernization program cannot succeed unless it understands:

Data Movement

How information travels.

Execution Flow

How processes execute.

Business Flow

How capabilities interact.

Transformation Flow

How modernization affects downstream systems.

Lineage Types

MIP supports four lineage categories.

```
1. Data Lineage

2. Execution Lineage

3. Business Lineage

4. Modernization Lineage
```

Data Lineage

Tracks movement of information.

Example:

```
CUSTOMER_FILE

↓
```

```
TS2LOAD

↓

CUSTOMER_MASTER

↓

TS2AUTH

↓

AUTHORIZATION_LOG

↓

SETTLEMENT_JOB
```

Questions Answered:

- Where did this field originate?
- What programs update it?
- What tables store it?
- Who consumes it?

Execution Lineage

Tracks runtime execution.

Example:

```
Scheduler

↓

Nightly Job

↓

Step 1

↓

Program A

↓
```

```
Program B

↓

Program C
```

Questions Answered:

- What starts execution?
- Which programs run next?
- Which batch chain is affected?

Business Lineage

Tracks business capability flow.

Example:

```
Card Activation

↓

Authorization

↓

Settlement

↓

Billing
```

Questions Answered:

- Which capability consumes outputs?
- Which domains interact?
- What business process is impacted?

Modernization Lineage

Tracks migration paths.

Example:

```
Authorization Capability

↓

Authorization Service

↓

API Layer

↓

Cloud Platform
```

Questions Answered:

- What replaces this component?
- What migration wave owns it?
- What future architecture depends on it?

Lineage Node Types

```
Application

Domain

Capability

Program

Job

Job Step

Transaction

Dataset

VSAM

DB2 Table

Column

API

Service Candidate
```

```
Migration Wave
```

Lineage Edge Types

Data Flow

```
READS

WRITES

PRODUCES

CONSUMES

UPDATES

TRANSFORMS
```

Execution Flow

```
CALLS

EXECUTES

STARTS

TRIGGERS

INVOKES
```

Business Flow

```
SUPPORTS

ENABLES

DEPENDS_ON
```

Modernization Flow

```
BECOMES

MIGRATES_TO

REPLACES
```

Canonical Lineage Path

Example:

```
Dataset

↓

Program

↓

Table

↓

Program

↓

Dataset

↓

Program

↓

Capability
```

This path becomes the foundation for:

- Impact Analysis
- Root Cause Analysis
- Modernization Planning

Data Lineage Model

Dataset Lineage

Example:

```
CUSTOMER_INPUT

↓

LOADJOB

↓

CUSTOMER_MASTER

↓

AUTHORIZATION

↓

AUTH_LOG
```

Store:

```
{
  "source": "CUSTOMER_INPUT",
  "target": "AUTH_LOG",
  "path_length": 4,
  "confidence": 0.95
}
```

Table Lineage

Example:

```
CARD_MASTER

↓

AUTH_PROGRAM

↓
```

```
SETTLEMENT_TABLE

↓

BILLING_TABLE
```

Questions:

- Which table feeds this table?
- What downstream tables depend on it?

Column Lineage

Future Enhancement

Example:

```
ACCOUNT_NUMBER

↓

CARD_MASTER.ACCOUNT_NUMBER

↓

AUTH_TABLE.ACCOUNT_NUMBER

↓

STATEMENT_TABLE.ACCOUNT_NUMBER
```

Critical for:

- Regulatory Reporting
- GDPR
- Compliance
- Audit

Execution Lineage Model

Batch Chain

Example:

```
JOB001

↓

TS2LOAD

↓

JOB002

↓

TS2AUTH

↓

JOB003

↓

TS2SETTLE
```

Questions:

- What batch chain depends on JOB001?
- Which downstream jobs are affected?

Online Transaction Flow

Example:

```
Transaction

↓

Program

↓

Program

↓

DB2

↓
```

```
Response
```

Questions:

- What path executes during authorization?
- Which systems participate?

Business Lineage Model

Maps technical assets to business value.

Example:

```
Program

↓

Capability

↓

Domain

↓

Application
```

Questions:

- Which capability owns this program?
- Which domain is impacted?

Lineage Construction Pipeline

```
Repository

↓

Metadata

↓

Relationships
```

```
↓

Knowledge Graph

↓

Lineage Builder

↓

Lineage Repository
```

Lineage Storage

SQLite Tables

```
lineage_node

lineage_edge

lineage_path

lineage_snapshot
```

Lineage Path Schema

```
CREATE TABLE lineage_path (

    path_id TEXT PRIMARY KEY,

    source_node TEXT,

    target_node TEXT,

    path_type TEXT,

    hop_count INTEGER,

    confidence_score REAL,

    created_at TIMESTAMP
);
```

NetworkX Lineage Construction

Use:

```
nx.MultiDiGraph()
```

Path Discovery:

```
nx.shortest_path()

nx.all_simple_paths()

nx.descendants()

nx.ancestors()
```

Core Lineage Algorithms

Upstream Lineage

Question:

```
Where did this originate?
```

Traversal:

```
Reverse Graph Traversal
```

Downstream Lineage

Question:

```
Who depends on this?
```

Traversal:

```
Forward Graph Traversal
```

End-to-End Lineage

Question:

```
How does this flow through the system?
```

Traversal:

```
Source → Target Path Analysis
```

Lineage Confidence Model

Every lineage path receives confidence.

Factors:

```
Parser Confidence

Metadata Quality

Relationship Confidence

Evidence Completeness
```

Explainability

Every lineage result includes:

```
{
  "path": [],
  "confidence": 0.94,
  "evidence": [],
  "reasoning": []
}
```

Example Questions

Data Origin

```
Where did ACCOUNT_NUMBER originate?
```

Downstream Impact

```
Who consumes CUSTOMER_MASTER?
```

Batch Dependency

```
Which jobs depend on JOB001?
```

Capability Impact

```
Which business capability depends on TS2AUTH?
```

Enterprise Scale Targets

Support:

```
100,000+ Programs

20,000+ Jobs

500,000+ Tables

10M+ Relationships

Millions of Lineage Paths
```

Success Criteria

A new engineer should be able to ask:

```
Show the complete lifecycle of a card authorization transaction.
```

And receive:

- Programs
- Jobs
- Tables
- Datasets
- Dependencies
- Capabilities
- Modernization Impact

without reading COBOL.

MIP Principle

Metadata captures facts.

Knowledge Graph captures relationships.

Lineage captures movement.

Impact Analysis captures consequences.

Modernization Intelligence captures opportunity.

End of Document.